



Phytoplankton responses to warming and eutrophication: evidence from a tropical mesocosm experiment

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Abstract Phytoplankton communities are sensitive to global change drivers, yet the combined effects of warming and nutrient enrichment remain insufficiently explored in tropical freshwater ecosystems. Here, we tested the individual and interactive effects of increased temperature and eutrophication on phytoplankton structure using outdoor mesocosms in a tropical region. We hypothesized that both stressors would influence community responses, promoting higher phytoplankton density, increased chlorophyll-*a*, reduced species richness, and shifts toward tolerant Chlorophyta and Cyanobacteria, with stronger

effects expected when warming and nutrient enrichment occurred together. We monitored limnological conditions and phytoplankton composition to capture short-term ecological responses. Nutrient enrichment triggered pronounced changes in water conditions, including higher turbidity, conductivity, dissolved oxygen, and chlorophyll-*a*, accompanied by clear shifts in phytoplankton density and species composition. Warming alone produced subtle effects but slightly reinforced eutrophication-driven increases in chlorophyll-*a* and influenced some limnological variables. Species richness remained stable across treatments, while Chlorophyta dominated under nutrient-rich conditions, reflecting their advantage in tropical eutrophic environments. Overall, eutrophication was the primary driver of phytoplankton responses, whereas warming acted as a secondary, modulatory factor. These findings highlight the heightened vulnerability of tropical freshwater ecosystems to nutrient enrichment and underscore the need to integrate nutrient reduction policies with climate change adaptation in freshwater management.

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Introduction

Freshwater ecosystems harbor exceptionally rich biodiversity, yet they are among the most threatened

environments on the planet (Dudgeon et al., 2006; Darwall et al., 2018). Multiple anthropogenic stressors, such as nutrient loading, land use change, habitat degradation, and contamination, have accelerated biodiversity loss in lakes and reservoirs worldwide (Dudgeon, 2019). Climate-driven changes, including atmospheric warming, altered precipitation regimes, and increased ultraviolet radiation, add further pressure to freshwater biota during the Anthropocene (Dudgeon, 2019; Ahmed et al., 2022; Teixeira et al., 2024). In addition, freshwater ecosystems are exposed to multiple anthropogenic stressors that frequently occur together, and their combined action can alter biodiversity patterns and ecosystem services in ways that are not captured when stressors are examined individually (e.g., Ormerod et al., 2010). It is important to emphasize that the effects of these stressors are not always linear, and their combined impacts can vary along stress gradients (e.g., Kefford et al., 2023). At low levels, some stressors may have neutral or even facilitative effects, whereas at higher intensities they may lead to strong negative responses or threshold shifts. Furthermore, interactions between stressors can be synergistic, additive, or antagonistic, making ecological predictions more complex (Piggott et al., 2015; Côté et al., 2016).

Among anthropogenic pressures, nutrient enrichment, particularly inputs of phosphorus and nitrogen from agricultural and urban sources, is one of the primary drivers of biodiversity decline in freshwater ecosystems (Dudgeon, 2014). Excessive nutrient loading often triggers eutrophication, leading to harmful algal blooms, oxygen depletion, shifts in trophic structure, and degradation of water quality (Dodds & Smith, 2016; Wurtsbaugh et al., 2019). At the same time, global warming is altering fundamental ecological processes in lakes, with consequences for species interactions, productivity, and biogeochemical cycles (Jeppesen et al., 2010). Understanding how these stressors interact is therefore critical for predicting ecosystem trajectories under future environmental change.

Phytoplankton, photosynthetic microorganisms that form the base of aquatic food webs, play a central role in these dynamics. As primary producers, they mediate trophic interactions (Velthuis et al., 2017), drive oxygen production and nutrient cycling (MEA, 2005; Naselli-Flores & Padisák, 2023), and respond rapidly to environmental fluctuations due

to their short generation times (Litchman & Klausmeier, 2008). Their community composition is governed by multiple drivers, including light availability, grazing, and concentrations of key nutrients such as phosphorus, nitrogen, iron, and silicate. Among these factors, temperature (Sunagawa et al., 2015) and nutrient availability, particularly phosphorus limitation (Redfield, 1958; Howarth & Marino, 2006), are widely recognized as major determinants of phytoplankton structure and productivity. Temperature influences metabolic rates, photosynthetic capacity, and growth (Daufresne et al., 2009; Chen et al., 2014; Nalley et al., 2018), while nutrient availability constrains biomass accumulation and cellular stoichiometry (Klausmeier et al., 2004; Marañón et al., 2012). Consequently, shifts in temperature and nutrient supply can cascade from individual-level physiological responses to community-level reorganization and changes in ecosystem functioning (Salmaso & Tolotti, 2021).

Despite the ecological importance of phytoplankton, experimental studies evaluating the combined effects of warming and eutrophication in freshwater systems remain relatively limited, particularly in tropical regions (e.g., Machado et al., 2023a). Recent studies have begun to address these interactions, highlighting changes in resource use efficiency and phytoplankton dynamics under different environmental scenarios (De Senerpont Domis et al., 2014; Anderson et al., 2022; Moresco et al., 2024; Nabout et al., 2025). In addition, broader ecological processes such as denitrification, trophic transfer, and nutrient cycling may also be influenced by the interaction between warming and eutrophication, especially in low-latitude systems. From a global change perspective, previous studies have emphasized that warming can intensify eutrophication effects by altering metabolic processes, species interactions, and ecosystem functioning (Burford et al., 2020; Huisman et al., 2018; Meerhoff et al., 2022). However, experimental evidence from tropical freshwater systems remains comparatively scarce, reinforcing the need for studies that explicitly address these combined stressors under tropical conditions.

In this context, mesocosm experiments offer a powerful framework for investigating how aquatic ecosystems respond to multiple anthropogenic stressors (e.g., Macaulay et al., 2025). By enabling the controlled manipulation of nutrients, temperature,

and other environmental drivers while maintaining ecological complexity, mesocosms allow researchers to evaluate mechanistic hypotheses and detect nonlinear or interactive effects that are difficult to observe in natural systems. Thus, the mesocosms provide a crucial bridge between theory and practice, linking controlled experimentation to real-world ecosystem functioning (e.g., Odum, 1984).

In this study, we conducted a mesocosm experiment in a tropical environment to evaluate the interactive effects of eutrophication and temperature increase on the phytoplankton community. Specifically, we aimed to: (i) assess how these stressors influence the composition, richness, and total density of phytoplankton species; and (ii) determine how species composition responds to the combined effects of elevated temperature and eutrophication. We also monitored limnological variables to better understand their behavior under the manipulated conditions. We hypothesized that both stressors together (e. g., temperature and nutrient enrichment) would alter the phytoplankton community by promoting higher population density, stimulating chlorophyll-*a* production, reducing species richness, and shifting species composition toward tolerant taxa. Based on previous studies, we predicted a reduction in species richness and a shift toward taxa tolerant to high temperature and nutrient availability, particularly chlorophytes and cyanobacteria, which are known to thrive under such conditions (Winder & Hunter, 2008; Machado et al., 2019, 2023b). Studies of this nature can provide valuable insights for the monitoring and management of aquatic ecosystems under scenarios of global anthropogenic change, particularly in tropical environments.

Material and methods

Experimental design

This study was conducted at the Experimental Area of Aquatic Ecology, located at the Universidade Estadual de Goiás (UEG, Anápolis, Goiás State, Brazil) (see Fig. S1 in the Supplementary Material SM1). The experiment was carried out over a 12-day period between November and December 2022, corresponding to the rainy season in the Brazilian Cerrado. The 12-day duration was chosen based on previous studies (e.g., Machado et al., 2023b), which showed that after

approximately 14–15 days, it becomes necessary to add nutrients to the control treatment due to nitrogen and phosphorus depletion. Here, we aimed to conduct the experiment without additional nutrient input, as nutrient enrichment could alter the dynamics of the control community.

We used a mesocosm system, which has proved to be an effective model for investigating ecological processes and responses in aquatic ecosystems under seminatural conditions (Spivak et al., 2011). A total of 30 mesocosms were established, each consisting of a polyethylene tank with an upper diameter of 1.24 m, a lower diameter of 1.00 m, a depth of 0.66 m, and a capacity of 500 L (Cobelo et al., 2023). The mesocosms were partially buried in the ground to minimize overheating and were exposed to ambient environmental conditions (Mesocosm, 2022; Machado et al., 2023b).

Water and the associated phytoplankton community used to fill the mesocosms was collected from an oligotrophic lake (total phosphorus = 0.013 mg L⁻¹; total nitrogen = 0.03 mg L⁻¹) located at the Experimental Station of the Agência Goiana de Assistência Técnica (EMATER, Anápolis, Goiás, Brazil; 16°20'27.5"S, 48°53'10.7"W). A suction hose was positioned near the center of the lake within the water column to minimize sediment resuspension and ensure the collection of the phytoplankton community. Water and associated biota were transferred using a suction pump connected to the hose into a tanker truck with an approximate capacity of 20,000 L. The water was not filtered prior to use.

Once transported to the experimental area, the 30 mesocosms were filled using a pump connected to a hose to ensure homogeneous distribution of the pond water. After filling, the phytoplankton community was allowed to acclimate for 24 h before the treatments were applied.

We simulated two key anthropogenic stressors: (i) eutrophication and (ii) temperature increase. For eutrophication, we manipulated nutrient concentrations to simulate two trophic states: eutrophic (E) with phosphorus levels typical of eutrophic ponds (e.g., Cunha et al., 2013) and oligotrophic (O) with phosphorus concentrations matching those of the source pond. Eutrophication was induced by adding phosphate (PO₄³⁻) in the form of potassium phosphate (KH₂PO₄). The concentrations followed the trophic state classifications reported in the literature

(e.g., Lamparelli, 2004; Cunha et al., 2013). Thus, eutrophic mesocosms received 0.120 mg L^{-1} of PO_4^{3-} , while oligotrophic mesocosms received 0.001 mg L^{-1} . These concentrations were applied equally to all mesocosms within each treatment at the beginning of the experiment. All mesocosms had equivalent nitrate (NO_3^-) concentrations of 1.92 mg L^{-1} , obtained from sodium nitrate (NaNO_3) solutions.

To simulate the temperature increase, a sheet of transparent polyethylene plastic (100 μm thick) was installed over the mesocosms, forming open chambers similar to those described by Netten et al. (2008) (see Fig. S2 in the Supplementary Material SM1). This type of plastic is commonly used in agricultural greenhouses and aquaculture systems because it allows the transmission of most of the incident solar radiation, retaining heat and creating an effect similar to that of a greenhouse that increases the internal temperature. Three temperature treatments were established: (i) ambient temperature (T_0 : $25.43 \pm 0.38^\circ\text{C}$, mean $\pm$ SD), representing the natural environmental temperature at the experimental site (Fig. S2A); (ii) moderate warming (T^+ : $27.97 \pm 0.48^\circ\text{C}$, mean $\pm$ SD), achieved using a plastic cover with a 40 cm central opening (Fig. S2B); and (iii) elevated warming (T^{++} : $28.73 \pm 0.35^\circ\text{C}$, mean $\pm$ SD), achieved using a cover with a 20 cm opening (Fig. S2C). Temperature was measured every two days using a multiparameter probe (Probest MP-301). The size of the opening determined the heating intensity, producing average increases of approximately $+2^\circ\text{C}$ and $+3^\circ\text{C}$ relative to the control, respectively. The openings on the plastic covers allowed for air exchange with the external environment, enabling some wind-driven mixing. In addition, the water in each mesocosm was manually mixed every two days with a PVC rod to prevent stratification and ensure water circulation. These temperature increments correspond to moderate (SSP1-1.9) and intermediate (SSP2-4.5) greenhouse gas emission scenarios projected by the Intergovernmental Panel on Climate Change (IPCC, 2021), representing expected warming trends for the twenty-first century.

Before applying the treatments, the phytoplankton community was homogenized across mesocosms by exchanging water using 5-L buckets, following the procedure described by Machado et al. (2023b). This exchange ensured that all units started the experiment with a similar species composition. It was the

only occasion in which water was transferred among mesocosms; throughout the remainder of the experiment, all units were maintained as isolated systems. After homogenization, the transparent covers were installed over the mesocosms assigned to the warming treatments, and nutrients were added according to the eutrophic or oligotrophic conditions, following a fully randomized experimental design. The experiment comprised six treatments, each with five replicates, totaling 30 mesocosms: (i) $E + T_0$; (ii) $E + T^+$; (iii) $E + T^{++}$; (iv) $O + T_0$; (v) $O + T^+$; and (vi) $O + T^{++}$ (see Fig. S3 in the Supplementary Material SM1).

Environmental and phytoplankton sampling

Limnological variables, including water temperature ($^\circ\text{C}$), pH, turbidity (NTU), conductivity ($\mu\text{S cm}^{-1}$), and dissolved oxygen (mg L^{-1}), were monitored every two days using a multiparameter probe (Probest MP-301), totaling seven sampling events. To estimate total phosphorus (TP; mg L^{-1}), orthophosphate (mg L^{-1}), ammoniacal N (mg L^{-1}), and nitrate (mg L^{-1}) concentrations throughout the experiment, 500 mL of water were collected from each of the 30 mesocosms at two sampling dates: day 0 (beginning of the experiment) and day 8, following an 8-day sampling interval. As the experiment lasted 12 days, the second sampling occurred four days before the end of the experiment. TP and orthophosphate concentrations were determined using the ascorbic acid–molybdate method (Golterman et al., 1978; method 4500-PE in APHA, 2005). Nutrient-related variables (phosphorus and nitrogen) were monitored to verify and maintain the intended trophic conditions, as these elements were experimentally manipulated. Therefore, they were not included as response variables in the statistical analyses, since they represent controlled components of the experimental design rather than independent drivers (see Fig. S4A–D in the Supplementary Material SM1).

Chlorophyll-*a* concentrations were measured every two days using the same multiparameter probe (Probest MP-301), resulting in seven sampling periods. To assess potential changes in trophic state, we applied the thresholds established by the Organization for Economic Co-operation and Development (OECD) (e.g., Gibson et al., 2000; Pantoja et al., 2021). Two main sampling points were selected for phytoplankton community analyses: the initial phase

of the experiment (day 0), when all mesocosms were untreated, and the onset of the chlorophyll-*a* peak (day 12). Before each community sampling event, the water column of each mesocosm was gently mixed using dedicated polyvinyl chloride (PVC) rods to avoid species loss by sedimentation, following Machado et al. (2019). Phytoplankton samples were then collected from the surface water using 100-mL amber glass bottles, immediately preserved with modified acetic Lugol's solution (Vollenweider et al., 1974), and stored until analysis.

Phytoplankton identification and counting were performed using 5-mL sedimentation chambers following the Utermöhl method (Utermöhl, 1958) under an inverted microscope (Zeiss Axio Vert. A1) at 400× magnification. Individuals (unicellular, colony and filaments forms) were counted in random fields until species accumulation curves stabilized. The density was estimated based on the number of individuals counted per field, the area of the chamber, and the volume of the settled sample, and expressed as individuals per milliliter (ind.mL^{-1}). Taxa were identified to the lowest possible taxonomic level (Bicudo & Menezes, 2017; Guiry & Guiry, 2023a). Density was expressed as individuals per milliliter (ind.mL^{-1}), and species richness was defined as the number of taxa per sample (Machado et al., 2023b).

Data analysis

We evaluated the effects of warming and eutrophication on limnological variables (pH, turbidity, conductivity, dissolved oxygen), chlorophyll-*a* concentration, total phytoplankton density, and taxon richness using a permutational repeated-measures ANOVA with 999 permutations. Temperature (three levels), trophic state (two levels), and time (sampling days) were included as fixed factors, and mesocosm identity was specified as the subject factor. All variables, except pH, were $\log(x+1)$ -transformed to reduce skewness and stabilize variances. When significant effects were detected, pairwise comparisons were made using post hoc tests with *P*-values adjusted by the Bonferroni method.

The effects of warming, eutrophication, and time on phytoplankton community composition were examined using a three-factor permutational multivariate analysis of variance (PERMANOVA; factors: temperature, trophic state, and time) based on

Bray–Curtis dissimilarities. Phytoplankton community data were expressed as species densities (ind.mL^{-1}) and Hellinger-transformed prior to analysis to reduce the influence of highly abundant taxa and to make the data suitable for multivariate analysis based on Bray–Curtis dissimilarities (Legendre & Gallagher, 2001). Before PERMANOVA, we tested the assumption of homogeneity of multivariate dispersion for each factor (temperature, trophic state, and time) using multivariate dispersion analysis. As a permutation-based approach was used, assumptions such as normality and sphericity are not required. Instead, inference relies on exchangeability under the null hypothesis, which was addressed through restricted permutations within mesocosms (Anderson & Braak, 2003). To visualize treatment effects and temporal trajectories in community structure, we performed a non-metric multidimensional scaling (NMDS) ordination based on Bray–Curtis distances. In addition, we assessed the relationships between individual taxa and the NMDS axes using vector fitting.

All statistical analyses were carried out in R (R Core Team 2023). Permutational repeated-measures ANOVAs were implemented with the *Permuco* package (Frossard & Renaud, 2019), post hoc tests with *rstatix* (Kassambara, 2021), and multivariate analyses (Hellinger transformation, Bray–Curtis dissimilarity, tests for homogeneity of dispersion, PERMANOVA, NMDS, and species–ordination relationships) with *vegan* (Oksanen et al., 2019). Data handling and graphics, including line plots and NMDS ordinations, were produced with *ggplot2* (Wickham, 2016).

Results

As expected, the manipulated variables, temperature and nutrient concentrations, showed clear and consistent differences among treatments throughout the experiment (Figs. S4, S5 in the Supplementary Material SM1). The temperature treatments effectively reproduced the targeted warming scenarios, with mean values of approximately 25.4 °C, 27.9 °C, and 28.7 °C for the ambient, moderate, and elevated conditions, respectively (see Fig. S5). Similarly, total phosphorus, orthophosphate, ammoniacal nitrogen, and nitrate concentrations remained higher in eutrophic treatments compared to oligotrophic ones,

confirming the maintenance of the intended trophic gradient (see Fig. S4).

Both the manipulated factors and time significantly influenced limnological dynamics (Table 1). However, trophic state was generally the main driver of variation, whereas temperature alone had comparatively minor effects. pH values varied significantly over time but remained within a neutral to slightly basic range throughout the experiment (Table 1; Fig. 1A). Turbidity showed moderate temporal fluctuations, and significant time–nutrient interactions indicated that eutrophic treatments tended to exhibit higher variability (Table 1; Fig. 1B).

Conductivity was consistently higher in eutrophic than in oligotrophic mesocosms across the experiment, with partial convergence among treatments near its end (Table 1; Fig. 1C). Dissolved oxygen concentrations also increased markedly under eutrophic conditions. The three-way interaction among temperature, nutrient enrichment, and time significantly explained variation in oxygen levels (Table 1). From day 4 onward, the eutrophic, non-heated treatment began to diverge from oligotrophic treatments, and by day 10, eutrophic mesocosms under moderate and elevated temperatures showed the highest oxygen concentrations (Fig. 1D).

Chlorophyll-*a* concentration showed a significant three-way interaction among temperature, nutrient enrichment, and time (Table 2). Eutrophic mesocosms exhibited a clear and progressive increase in chlorophyll-*a* concentration throughout the experiment, whereas oligotrophic mesocosms remained consistently low across all temperature treatments (Fig. 2). From day 4 onward, eutrophic treatments under ambient and moderate temperatures already showed higher chlorophyll-*a* concentrations than all

oligotrophic treatments. By the end of the experiment (days 10–12), eutrophic mesocosms under moderate ($E + T^+$) and elevated ($E + T^{++}$) temperatures reached chlorophyll-*a* levels characteristic of eutrophic to hypereutrophic conditions according to OECD thresholds, while oligotrophic mesocosms remained within the oligotrophic to mesotrophic range (Fig. 2).

A total of 133 phytoplankton taxa were recorded, distributed across 11 taxonomic classes. The most species-rich group was Chlorophyceae (77 taxa), followed by Cyanobacteria (14), Zygnematomyceae (9), Chrysophyceae (6), Cryptophyceae (5), Dinophyceae (5), Bacillariophyceae (5), Trebouxiophyceae (6), Klebsormidiophyceae (3), Euglenophyceae (2), and Raphidiophyceae (1).

Total phytoplankton density was significantly affected by nutrient enrichment and sampling time (Table 2). At the beginning of the experiment, all mesocosms exhibited similar total densities, but differences became evident as the experiment progressed. By the end of the experiment, eutrophic mesocosms showed markedly higher total cell densities than oligotrophic ones (Fig. 3A). The two green algal species *Chlamydomonas* sp. and *Chloromonas* sp. were the main contributors to this increase, representing 28.2% and 43.0% of the total density in eutrophic treatments, respectively. Species richness was not significantly influenced by temperature, nutrient enrichment, or time, and no significant interactions were detected among these factors (Table 2). Species richness values remained relatively stable across treatments, with overlapping trends between oligotrophic and eutrophic mesocosms (Fig. 3B).

The density of small cyanobacterial taxa (e.g., *Cyanobium plancticum* (G. Drews, H. Prauser & D. Uhlmann) Komárek, J. Kopecký & Cepák;

Table 1 *F* and *P* values from the permutational ANOVA performed for limnological variables, testing the effects of temperature increase, nutrient enrichment, and sampling time. Significant factors are highlighted in bold ($P < 0.05$).

	pH		Turbidity		Conductivity		Dissolved oxygen	
	<i>F</i>	<i>P</i>	<i>F</i>	<i>P</i>	<i>F</i>	<i>P</i>	<i>F</i>	<i>P</i>
Temperature	0.934	0.424	0.276	0.765	2.956	0.050	0.516	0.610
Trophic	7.674	0.012	5.800	0.029	228.368	0.001	276.133	0.001
Temperature:Trophic	0.801	0.455	0.606	0.527	3.021	0.054	8.100	0.001
Time	91.387	0.001	9.078	0.001	73.358	0.001	68.069	0.001
Temperature:Time	1.048	0.404	0.734	0.739	1.150	0.316	2.645	0.001
Trophic:Time	3.200	0.006	4.340	0.001	15.434	0.001	36.032	0.001
Temperature:Trophic:Time	1.267	0.239	0.663	0.776	0.718	0.750	2.269	0.008

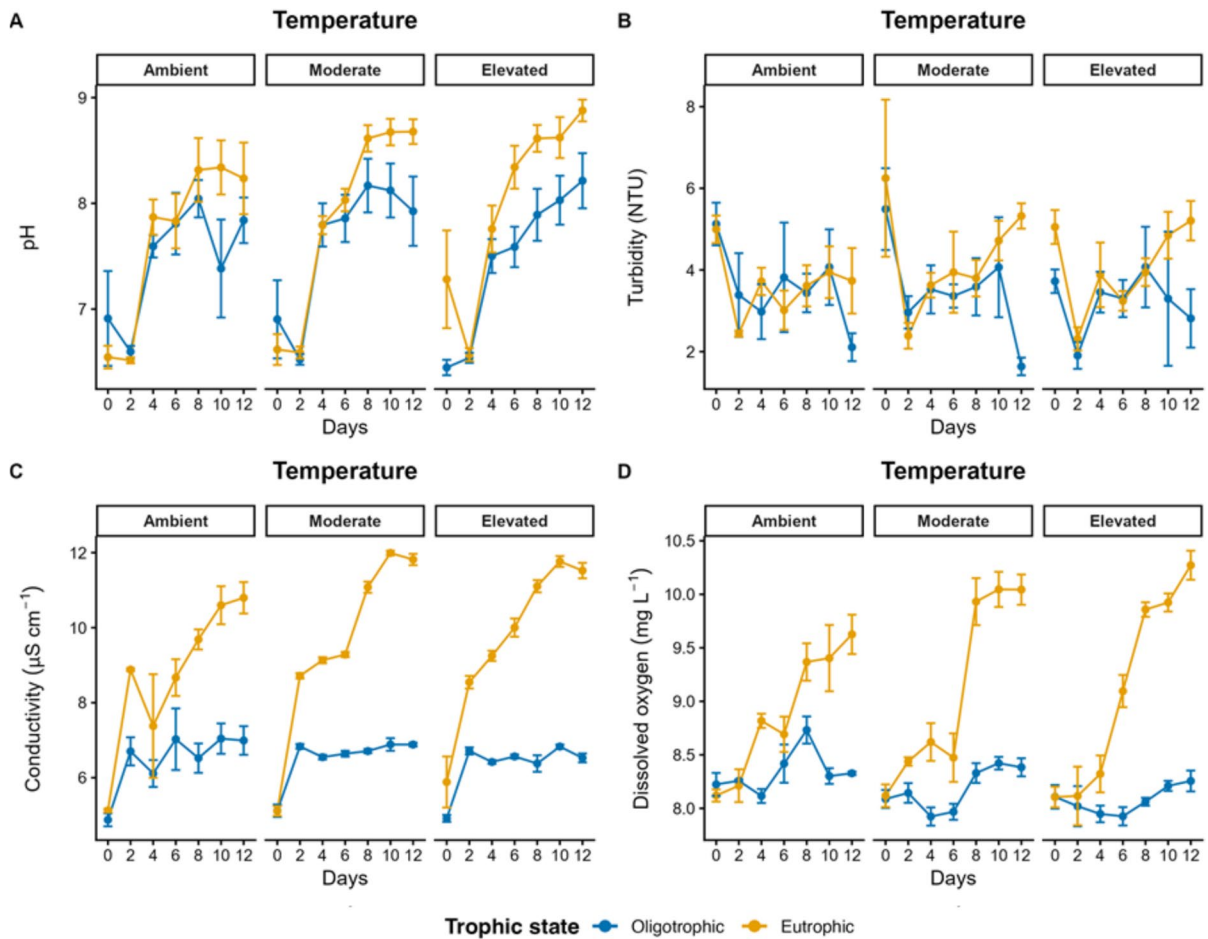


Fig. 1 Temporal variation in limnological variables measured under three temperature treatments, ambient; moderate (+2 °C), and elevated (+3 °C), for oligotrophic (blue) and

eutrophic (orange) mesocosms. (A) pH, (B) turbidity (NTU), (C) electrical conductivity ($\mu\text{S cm}^{-1}$), and (D) dissolved oxygen (mg L^{-1}). Error bars represent standard error of the mean

Table 2 *F* and *P* values from the permutational ANOVA performed for chlorophyll-*a*, total phytoplankton density and taxon richness, testing the effects of temperature increase, trophic state, and time. Significant factors are highlighted in bold ($P < 0.05$).

	Chlorophyll- <i>a</i>		Total density		Species richness	
	<i>F</i>	<i>P</i>	<i>F</i>	<i>P</i>	<i>F</i>	<i>P</i>
Temperature	27.651	0.001	0.556	0.620	3.519	0.052
Trophic	831.840	0.001	48.701	0.001	2.471	0.139
Temperature:Trophic	1.926	0.159	2.045	0.136	1.608	0.235
Time	266.631	0.001	253.111	0.001	0.091	0.762
Temperature:Time	4.288	0.001	0.269	0.787	0.339	0.711
Trophic:Time	84.570	0.001	53.188	0.001	0.262	0.599
Temperature:Trophic:Time	1.299	0.224	2.427	0.100	0.038	0.972

Cyanodictyon sp. and *Synechocystis* sp.) varied considerably among treatments and over time (Fig. 4). *Cyanobium plancticum* showed high temporal

variability, with low densities at day 0 and occasional peaks at day 12, particularly under oligotrophic conditions and elevated temperature. *Cyanodictyon* sp.

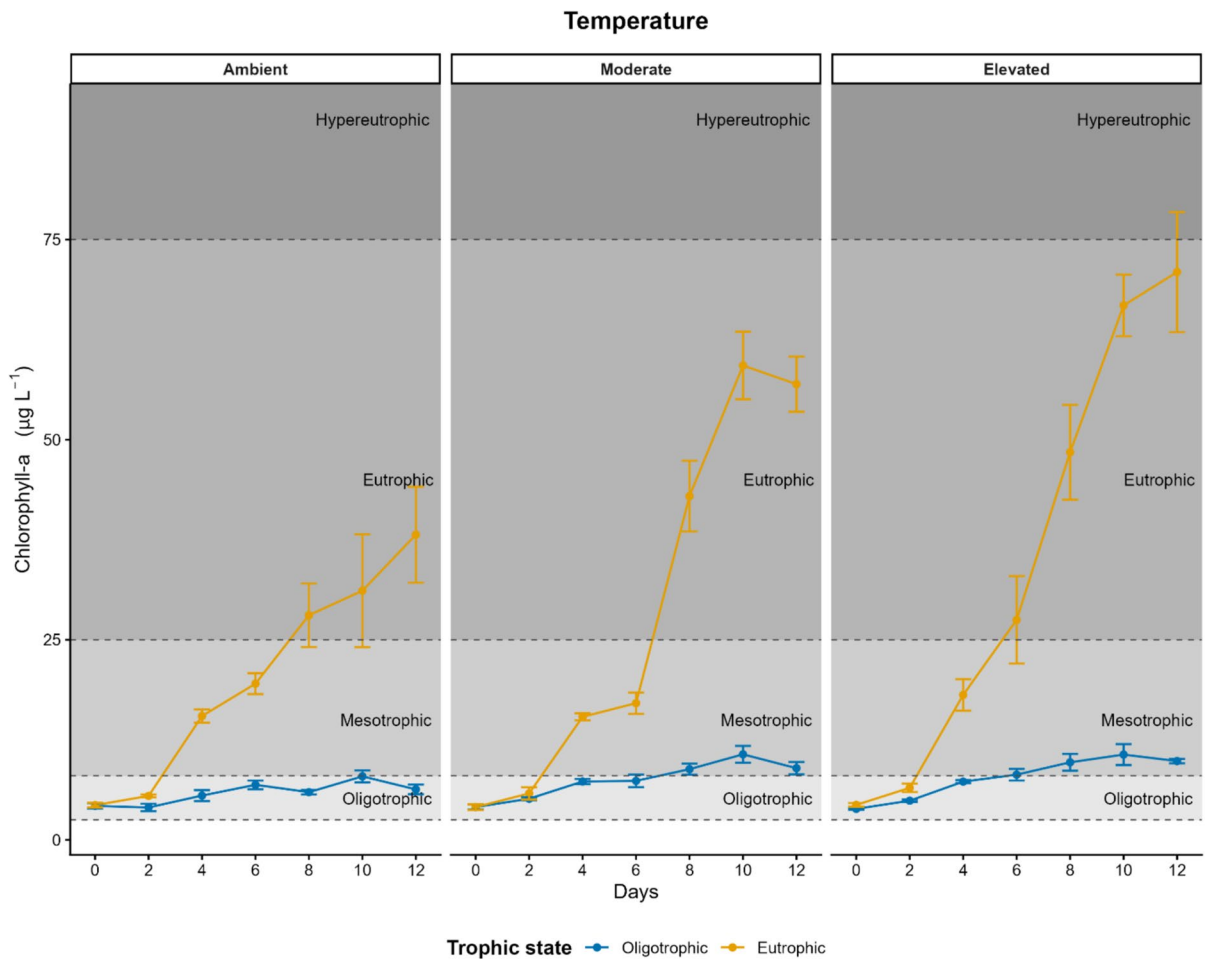


Fig. 2 Temporal variation in chlorophyll-*a* concentration ($\mu\text{g L}^{-1}$) under three temperature treatments, ambient; moderate (+2 °C); and elevated (+3 °C), for oligotrophic (blue) and eutrophic (orange) mesocosms. Shaded bands represent trophic

state categories based on chlorophyll-*a* thresholds proposed by OECD (see material and methods). Error bars represent the standard error of the mean ($n=5$)

showed a decline in density over time, with values close to zero at day 12 in most treatments, regardless of temperature and nutrient conditions. *Synechocystis* sp. showed a consistent increase in density from day 0 to day 12 under all temperature and nutrient conditions, with higher values under moderate and elevated temperatures, especially in eutrophic treatments.

Community composition was significantly influenced by trophic state and sampling time (PERMANOVA; Trophic State: $F=8.085$, $P=0.001$, $R^2=0.07$; Time: $F=29.754$, $P=0.001$, $R^2=0.28$; Trophic State $\times$ Time: $F=9.441$, $P=0.001$, $R^2=0.09$). The NMDS ordination indicated a strong temporal gradient on the first axis, separating the

initial (day 0) samples from the final (day 12) samples, irrespective of temperature treatment (stress = 0.126). Along the second axis, trophic state structured the final communities: Eutrophic mesocosms plotted at positive scores and oligotrophic mesocosms at negative scores, with no clear segregation by temperature within each trophic state (Fig. 5).

Species associations were consistent with the overall NMDS patterns. The first axis represented the temporal gradient of the experiment: samples on the negative side corresponded to the initial phase (with all temperature and trophic treatments overlapping), while those on the positive side corresponded to the final phase, indicating a marked shift in community

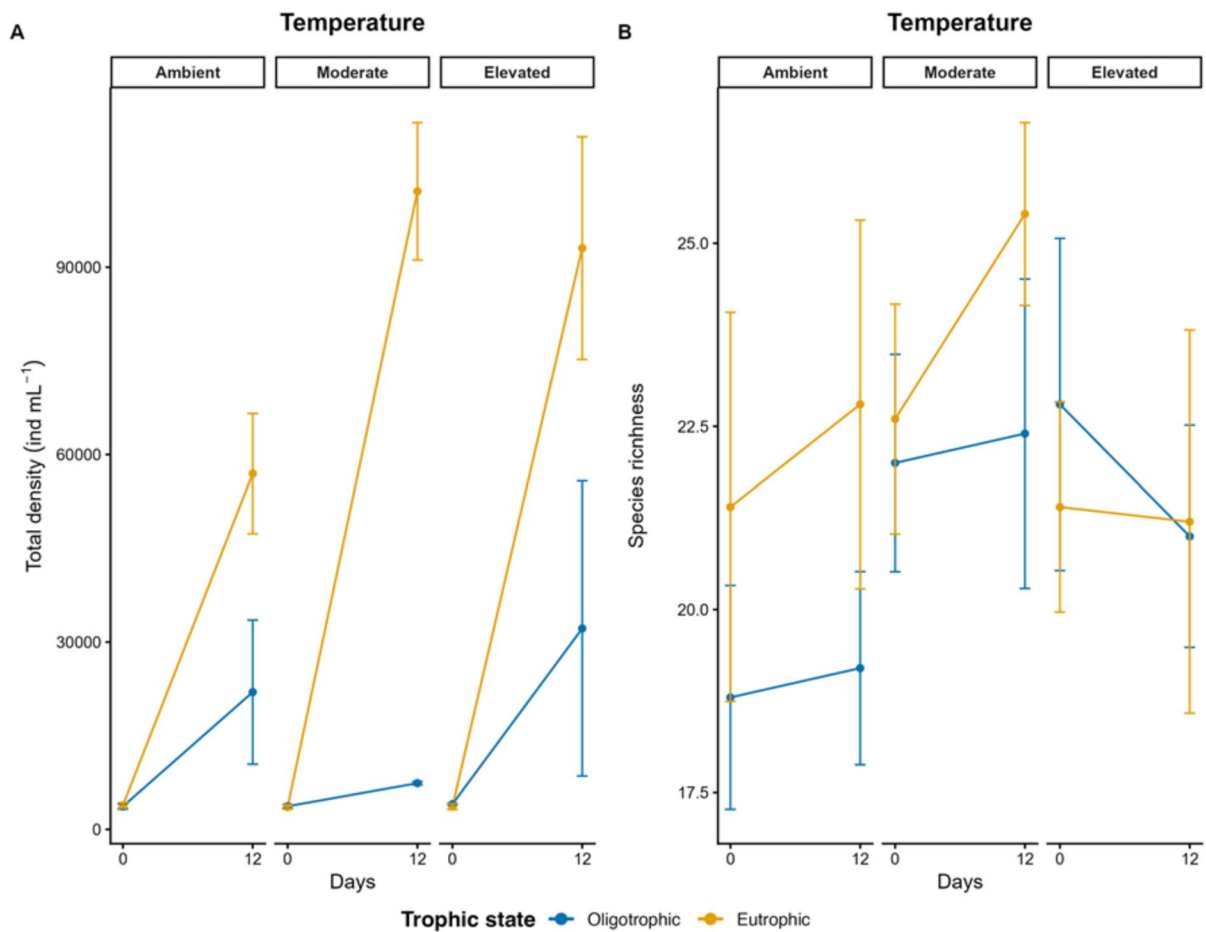


Fig. 3 Temporal variation in (A) total phytoplankton density (ind.mL⁻¹) and (B) species richness under three temperature treatments, ambient; moderate and elevated, for oligotrophic

(blue) and eutrophic (orange) mesocosms. Error bars represent the standard error of the mean

composition over time. Taxa located at the extremes of the NMDS axes were interpreted as being associated with the main experimental gradients (time along axis 1 and trophic state along axis 2). Fifty-five taxa contributed most strongly to the positive side of axis 1, including *Coelastrum* sp. 2, *Golenkinia radiata* Chodat, *Ankistrodesmus* sp. 1, and *Pandorina* sp. 1, mainly composed of chlorophytes. Conversely, seventy-eight taxa were associated with the negative side of axis 1, particularly *Planktothrix* sp. 1, *Schizochlamydes* sp. 1, *Desmodesmus* sp. 1, and *Pinnularia* cf. *alpina*, indicating community replacement at the end of the experiment. The second axis separated the trophic treatments: The positive side was associated with eutrophic mesocosms, characterized by *Coelastrum* sp. 1, *Aphanocapsa elachista* West

& GSWest, and *Aphanocapsa delicatissima* West & GSWest, whereas the negative side corresponded to oligotrophic mesocosms, dominated by *Cosmarium subcostatum* Nordstedt, *Desmodesmus abundans* (Kirchner) EHHegewald, and *Peridinium* sp. 2. Detailed relationships between taxa and NMDS ordination axes are presented in Supplementary Material SM2.

Discussion

This study integrated two major environmental stressors, nutrient enrichment and temperature increase, to evaluate their combined effects on phytoplankton communities. We found that eutrophication was the

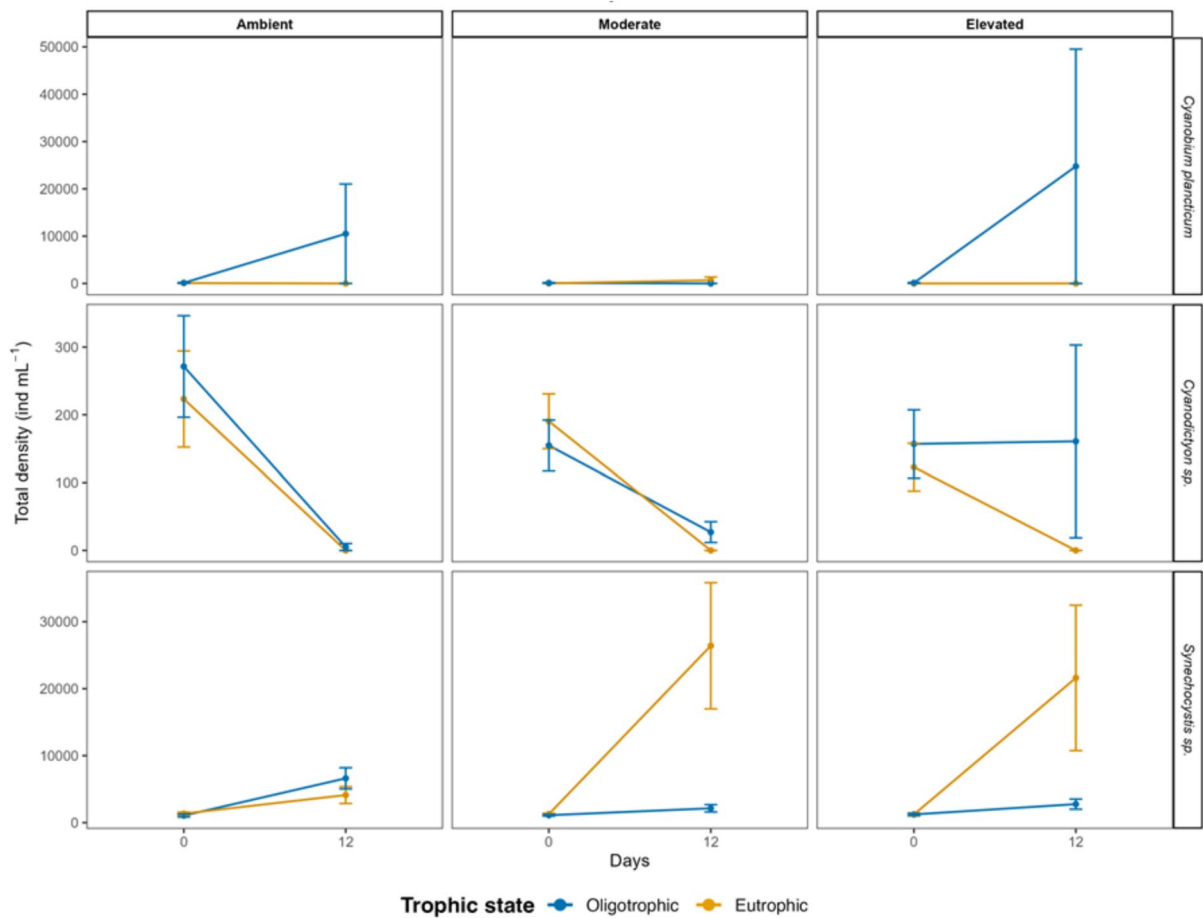


Fig. 4 Temporal variation in the density of small cyanobacteria taxa (ind.mL⁻¹) under three temperature treatments: ambient, moderate, and high, for oligotrophic (blue) and eutrophic (orange) mesocosms. Error bars represent the standard error of the mean

main factor driving short-term changes in phytoplankton structure and species composition, leading to distinct assemblages between oligotrophic and eutrophic conditions. Temperature, in turn, showed no significant effect when acting alone but interacted with nutrient enrichment to enhance chlorophyll-*a* concentrations. Overall, our hypotheses were partially supported, indicating that elevated nutrient levels can rapidly alter phytoplankton composition and chlorophyll-*a*, with potential consequences for the functioning and stability of aquatic ecosystems, particularly in tropical freshwater systems, where high baseline temperatures and elevated metabolic rates are likely to amplify phytoplankton responses to nutrient enrichment.

Consistent with our expectations and previous studies (e.g., Costa et al., 2018; Zhou et al., 2020;

Beentjes et al., 2022; Meerhoff et al., 2022; Lin et al., 2023; Machado et al., 2023a; Nabout et al., 2025), eutrophication was the main driver of changes in phytoplankton biomass and community composition, promoting the rapid proliferation of chlorophytes and cyanobacteria and leading to marked shifts in total density and community structure within a few days. Increased nutrient availability can accelerate growth rates and favor fast-growing, opportunistic taxa with high resource uptake efficiency (Smith, 2016; Dittrich et al., 2023). As a consequence, this process can promote rapid species selection and competitive dominance, resulting in fast changes in community structure (Reynolds, 2006).

In tropical eutrophic freshwater systems, chlorophytes and cyanobacteria often dominate phytoplankton communities, particularly under high

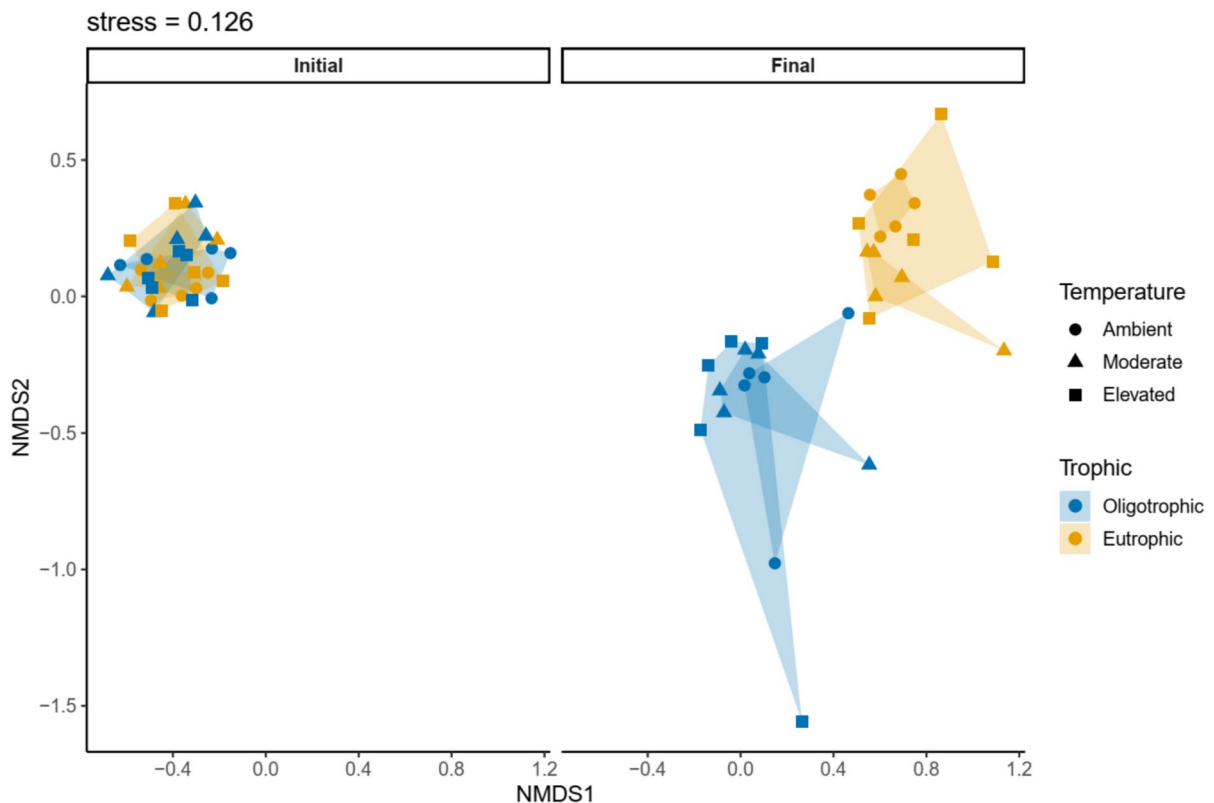


Fig. 5 Non-metric multidimensional scaling (NMDS) ordination of phytoplankton community composition based on Hellinger-transformed data and Bray–Curtis distances (stress=0.126). Facets represent sampling times (initial and

final). Points correspond to individual mesocosms, colored by trophic state (blue=oligotrophic, orange=eutrophic) and shaped by temperature treatment (ambient, moderate, elevated)

nutrient availability (Amorim & Nascimento 2021). In our experiment, *Chlamydomonas* sp. 1 and *Chloromonas* sp. 1, both chlorophytes commonly found in nutrient-rich systems, were the main contributors to total community density. These genera are typically characterized by fast growth rates, high physiological plasticity, and efficient nutrient uptake (Salomé & Merchant, 2019; Guiry & Guiry, 2023b), which allow them to rapidly exploit increased resource availability under eutrophic conditions. Their dominance likely reflects the high nutrient availability, particularly of phosphorus and nitrogen, which favors chlorophyte growth (Zhu et al., 2010), indicating strong environmental filtering driven by nutrient enrichment. An additional indication of temporal community replacement was the disappearance or decline of Bacillariophyceae and Chrysophyceae, which were initially present at day 0 but became rare or absent by the end of the experiment, likely due to their lower competitive

ability under high nutrient availability and the rapid dominance of fast-growing chlorophytes. This pattern should be interpreted with caution. In particular, diatoms, which are often favored under well-mixed conditions (Jäger et al., 2008), may have been disadvantaged by the reduced turbulence in the mesocosms, which can limit their ability to remain suspended in the water column. Therefore, the observed decrease in Bacillariophyceae likely reflects a combination of experimental conditions (e.g., limited mixing) and biotic responses to nutrient enrichment.

Cyanobacterial taxa may exhibit variable responses to nutrient and temperature effects (Huisman et al., 2018). The observed variation in the density of small cyanobacteria among treatments and over time suggests that these organisms respond sensitively to changes in temperature and nutrient availability, likely reflecting their small cell size and high nutrient uptake efficiency (Paerl & Huisman, 2008; Beaulieu

et al., 2013). The species *Cyanobium plancticum* showed high temporal variability, with peaks under oligotrophic conditions and high temperatures, reflecting an opportunistic strategy and efficient use of resources in nutrient-poor environments, as is typical of the cyanobacteria group (Dignum et al., 2005; Cottingham et al., 2015), likely enhanced by increased metabolic rates under warming (Kosten et al., 2012). In contrast, *Cyanodictyon* sp. showed a decline in density over time across all treatments, suggesting lower tolerance or competitive ability under variable conditions, potentially due to competitive exclusion by faster-growing taxa such as chlorophytes (e.g., *Chlamydomonas* sp. 1 and *Chloromonas* sp. 1), which dominated eutrophic mesocosms both with and without temperature increase. In turn, *Synechocystis* sp. increased consistently, particularly under moderate and elevated temperatures and eutrophic conditions, indicating high physiological plasticity and the synergistic effects of warming and nutrient enrichment on growth (Fernández-González et al., 2020). Although cyanobacteria generally exhibit lower growth rates compared to other phytoplankton groups, these patterns highlight taxon-specific strategies among small cyanobacteria and suggest that warming, and especially eutrophication, may shift community composition toward opportunistic taxa, with important implications for freshwater ecosystem functioning (Kosten et al., 2012; Yan et al., 2017).

Although phytoplankton organisms typically have short life cycles and can respond rapidly to environmental changes (Reynolds, 2006), the 12-day duration of this experiment primarily captured early-stage or transient dynamics rather than long-term successional processes. While fast-growing taxa (e.g., Chlorophyceae) may rapidly increase in abundance, shifts in community dominance, the establishment of slower-growing groups (e.g., Cyanobacteria), and changes in functional traits or biomass allocation often require longer time periods to become fully expressed, as evidenced by long-term studies showing gradual community shifts (e.g., Abonyi et al., 2018). Therefore, our results should be interpreted as reflecting short-term responses, and longer experiments would be necessary to capture more complete successional trajectories.

Although temperature increase is recognized as a major global change driver, its effects on phytoplankton composition are often weaker than those

of nutrient enrichment. In our experiment, warming alone did not significantly alter species richness or community structure. Importantly, the absence of significant changes in species richness does not imply a lack of community response, but rather indicates that environmental filtering primarily affected species identity and dominance patterns rather than richness per se. This interpretation is supported by previous studies suggesting that moderate increases in temperature tend to influence physiological rates, biomass, or cell size more strongly than taxonomic turnover (Winder & Sommer, 2012; Zohary et al., 2021). This pattern is also consistent with findings from DNA metabarcoding studies in tropical mesocosms (Nabout et al., 2025), where warming had limited independent effects on microbial composition compared with nutrient addition (e.g., Machado et al., 2020). The temperature increments applied (+2 to +3 °C) remain within the optimal thermal range for many phytoplankton species (Thomas et al., 2016), which may explain the absence of strong compositional responses. It is important to note, however, that temperature rise can influence algal physiology and functional traits at the population level, such as reducing reproductive rates or decreasing cell size (Yvon-Durocher et al., 2011; Sommer et al., 2015), and such changes may only translate into detectable community-level responses over longer experimental periods. We recognize that metrics such as phytoplankton community biovolume can act as an important biological indicator to describe the response of this group to the stressors we tested (Baho et al., 2017; Moresco et al., 2024). Therefore, longer-term mesocosm experiments incorporating biovolume and functional trait approaches may be necessary to fully understand how warming interacts with nutrient enrichment to shape phytoplankton community trajectories.

The combination of warming and nutrient enrichment intensified several limnological responses, including dissolved oxygen, turbidity, and chlorophyll-*a*. This is consistent with experimental evidence suggesting that temperature can amplify the impacts of eutrophication under high nutrient availability (Hennemann & Petrucio, 2010; Woolway et al., 2020), as higher temperatures enhance metabolic activity and nutrient uptake, leading to greater biomass accumulation (Moresco et al., 2024), which in turn increases turbidity and alters oxygen dynamics.

Although warming did not independently alter phytoplankton composition in our study, its interaction with eutrophication may have ecological relevance over longer timescales, potentially reinforcing bloom dynamics and favoring tolerant taxa.

Limnological variables can be directly or indirectly affected by eutrophication and rising temperatures. Previous studies have reported reductions in pH (McKee et al., 2003) and increases in conductivity under warming (McKee et al., 2003; Feuchtmayr et al., 2009) and eutrophication scenarios (Machado et al., 2023b), as these stressors influence key biogeochemical processes in aquatic systems. In our experiment, pH was affected exclusively by time, remaining within a neutral to slightly basic range. Conversely, conductivity responded to the interaction among temperature, nutrient enrichment, and time. This interaction likely amplified conductivity over time, as higher temperatures can enhance ion mobility and accelerate the release of dissolved ions into the water column, leading to increased conductivity (Tundisi & Tundisi, 2008).

Changes in dissolved oxygen due to warming have been observed in both directions across studies, with decreases (Feuchtmayr et al., 2009) or increases (Hennemann & Petrucio, 2010). Although nutrient enrichment is often associated with oxygen depletion (Misra & Chaturvedi, 2016), our results revealed a positive interaction between temperature increase ($T^+ = 27.4 \pm 0.41$ °C; $T^{++} = 28.2 \pm 0.35$ °C) and eutrophication on dissolved oxygen concentrations. This increase coincided with rising chlorophyll-*a* levels, suggesting that enhanced photosynthetic activity contributed to oxygen production (Janzen et al., 2008). Turbidity also increased as a result of the interaction among time, warming, and nutrient enrichment, mirroring the chlorophyll-*a* pattern and consistent with Machado et al. (2023b). Such correspondence likely reflects increased phytoplankton density in enriched treatments.

Although this was a short-term experiment, our results revealed clear and rapid responses of phytoplankton communities to eutrophication, highlighting their sensitivity and suitability as model systems for studying multiple stressors. Moreover, the tropical freshwater ecosystems are particularly sensitive to the combined effects of eutrophication and warming, making our results especially relevant in this context. High baseline temperatures, intense solar radiation,

and naturally accelerated metabolic rates in tropical systems can amplify the speed and magnitude of phytoplankton responses to nutrient enrichment, leading to rapid shifts in biomass, community structure, and water quality conditions (Heinrichs et al., 2025). Moreover, tropical regions such as the Brazilian Cerrado are experiencing increasing climatic instability, including more frequent and intense droughts (e.g., Teixeira et al., 2024), which reduce water volume and further concentrate nutrients, thereby intensifying eutrophication impacts. Despite these vulnerabilities, experimental assessments of multiple stressors in tropical mesocosms remain relatively scarce compared with temperate regions (e.g., Macaulay et al., 2025), limiting our ability to generalize ecological responses under future climate scenarios. Our findings therefore contribute valuable evidence that phytoplankton communities in tropical systems can undergo rapid restructuring under nutrient enrichment, and that even moderate warming can modulate these effects, emphasizing the need for expanded experimental research and targeted management strategies in tropical freshwaters.

Concluding remarks

Overall, our initial hypotheses were only partially supported. We observed clear increases in phytoplankton biomass and a marked dominance of chlorophytes, but this pattern did not extend strongly to cyanobacteria as initially predicted. However, when small cyanobacterial taxa were analyzed separately, we observed increases in density, particularly for *Synechocystis* sp., under eutrophic conditions combined with moderate and elevated temperatures. Chlorophyte dominance was driven primarily by nutrient enrichment, with little direct influence of warming or its interaction with trophic state. This likely reflects the fact that the temperature increments applied in this study remained within the physiological tolerance of many tropical phytoplankton taxa. Nevertheless, the combination of elevated temperature and eutrophication did produce measurable effects on ecosystem functioning, including significant increases in chlorophyll-*a* and consistent changes in limnological variables such as turbidity, conductivity, and dissolved oxygen. These results indicate that even moderate warming can

accentuate some of the impacts of eutrophication, despite having limited effects on community structure when acting alone.

Importantly, our findings highlight the sensitivity of tropical freshwater phytoplankton to nutrient enrichment and underscore the potential for rapid ecological change in these environments. Tropical systems, characterized by high baseline temperatures, intense solar radiation, and naturally accelerated metabolic rates, may respond more quickly and strongly to nutrient loading than their temperate counterparts. The interactive effects of warming and eutrophication therefore warrant particular attention in tropical regions, where climate-driven increases in temperature and hydrological instability may exacerbate eutrophication risks.

Taken together, our results reinforce the pivotal role of nutrient enrichment as a driver of phytoplankton community restructuring and demonstrate that warming, although not a primary driver of taxonomic change, can amplify eutrophication effects on ecosystem functioning. These findings emphasize the urgent need to monitor and manage nutrient inputs in tropical freshwaters, especially under ongoing climate warming, to prevent the rapid onset of undesirable ecological states and safeguard water quality.

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Author contributions ATA, MFA, KBM, and JCN were involved in conceptualization, methodology, and writing—reviewing and editing. ATA and MFA performed the experiment, did the sampling, and identified the biological data. ATA and JCN were responsible for formal analysis and investigation and writing—original draft preparation. JCN acquired the funding. KBM and JCN carried out supervision.

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Data availability Data are available from the authors upon reasonable request.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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